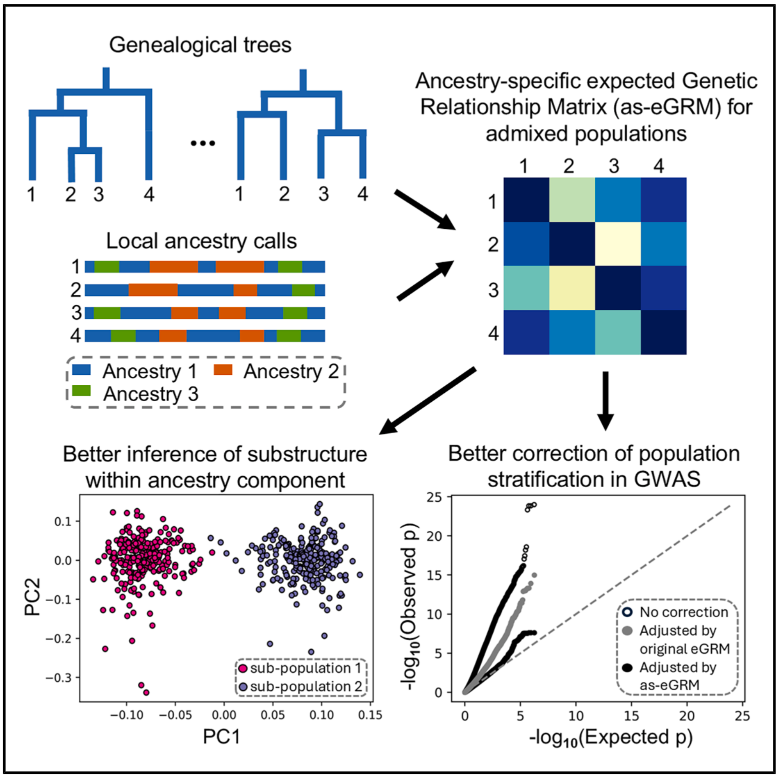
A genealogy-based approach for revealing ancestry-specific structures in admixed populations

Graphical abstract



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We present a method that combines inferred gene genealogical trees with local ancestry calls to estimate ancestry-specific expected genetic relationship matrix (as-eGRM) for admixed populations. Extensive evaluation shows that as-eGRM improves the inference of fine-scale substructure within ancestry components and the correction of population stratification in genome-wide association studies.

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ARTICLE

A genealogy-based approach for revealing ancestry-specific structures in admixed populations

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Summary

Elucidating ancestry-specific structures in admixed populations is crucial for comprehending population history and mitigating confounding effects in genome-wide association studies. Existing methods to reveal the ancestry-specific structures generally rely on frequency-based estimates of genetic relationship matrix (GRM) among admixed individuals after masking segments from ancestry components not being targeted for investigation. However, these approaches disregard linkage information between markers, potentially limiting their resolution in revealing structure within an ancestry component. We introduce ancestry-specific expected GRM (as-eGRM), a novel framework for estimating the relatedness within ancestry components between admixed individuals. The key design of as-eGRM consists of defining ancestry-specific pairwise relatedness between individuals based on genealogical trees encoded in the ancestral recombination graph (ARG) and local ancestry calls and then computing the expectation of the ancestry-specific relatedness across the genome. Comprehensive evaluations using both simulated stepping-stone models of population structure and empirical datasets based on three-way admixed Latino cohorts showed that analysis based on as-eGRM robustly outperforms existing methods in revealing the structure in admixed populations with diverse demographic histories, which in turn improves the robustness against confounding due to population structure in association testing.

Introduction

Genetic admixture, the exchange of genetic material of previously relatively isolated populations, results in haplotypes descended from multiple ancestral sources.[1–3] This phenomenon is common in human populations, exemplified by the genetic admixture experienced by native populations throughout the American continent due to the colonization by Europeans and the subsequent African slave trade.[4,5] Revealing ancestry-specific structures in admixed populations is crucial for understanding population history and adjusting for population stratification in genome-wide association studies (GWASs). These structures provide insights into migration patterns and genetic diversity, improving our understanding of complex population histories.[4,6] In GWASs, failure to account for population structure can lead to spurious associations or mask genuine genetic effects.[7–9] However, inferring these structures presents significant challenges due to the intricate genetic composition of admixed individuals, particularly in cases of recent admixture or populations with multiple ancestral sources.

The conventional approach for revealing population structure involves constructing a variance-standardized genetic relationship matrix (GRM) and applying principal-component analysis (PCA), sometimes in conjunction with uniform manifold approximation and projection (UMAP), to the GRM.[10–17] In the context of admixed populations, these approaches effectively average over the distribution of ancestral background at a genetic variant and across all loci in the genome, without incorporating ancestry information. Consequently, multiple components of ancestries could mask the finer-scale structure that may be of interest as inter-continental distances tend to dominate and explain the largest amount of variation in the GRM. Therefore, PCA or UMAP applied directly to GRM from admixed individuals tend to reveal structure driven by different proportions of ancestries, even among the lower principal components (PCs).

To address this limitation, Moreno-Estrada et al.[4] proposed an ancestry-specific PCA method named ASPCA. ASPCA masks genomic components derived from non-target ancestral populations and then compute the subspace spanned by the first several PCs of interest by finding a matrix decomposition that minimizes the reconstruction error.[4,18] After observing artifactual separation of clusters between reference and admixed individuals when using ASPCA, Browning et al.[6] proposed a variant of this ancestry-specific PCA method (we refer to this method as Browning's ancestry-specific multidimensional scaling [AS-MDS]), which applies MDS to a Euclidean distance matrix based on pairwise allelic differences between individuals after non-target ancestries are similarly masked. Finally, though not yet peer-reviewed, another ancestry-specific PCA method (missing DNA PCA [mdPCA]; see web resources) is also available, which constructs a covariance matrix that masks the components with non-target ancestries and then utilizes multiple matrix denoising techniques and truncated singular value decomposition (SVD) on the covariance matrix to compute ancestry-specific PCs. In all these methods linkage information was discarded, and thus these methods

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are expected to not fully utilize the genomic information when inferring population structure.

The entire genealogy of the DNA sequence of a sample of individuals can be represented by a series of genealogical trees connected through recombination events, collectively known as the ancestral recombination graph (ARG).[19,20] With the recent ability to infer or approximate the ARG in thousands of individuals, multiple downstream ARG-based population and statistical genetic applications have been developed to enhance our understanding of the evolutionary history of a population.[21–24] We previously developed an ARG-based framework, called eGRM, to infer the expected relatedness between pairs of individuals.[25] eGRM utilizes the same variance-standardized framework as the canonical GRM but sums over the vector of haploid individuals for each branch, weighted by branch lengths. As this approach leverages haplotype information that was used to infer the ARG, it enhances robustness when working with incomplete genetic data and improves over canonical GRM in elucidating the population structure of a sample through PCA and UMAP. However, eGRM does not remove the components with non-target ancestries, limiting its application to detect ancestry-specific structure in admixed populations.

In this study, we propose as-eGRM, a framework that integrates ARGs and local ancestry information to infer the expectation of pairwise genetic relatedness within ancestries in an admixed population. We show that PCA and UMAP applied to as-eGRM can outperform alternative methods such as AS-MDS and mdPCA in revealing ancestry-specific structures in admixed populations. We used simulated data of varying complexity to extensively evaluate the performance of as-eGRM on revealing the finer structure in admixed populations. We then applied as-eGRM to a real-world dataset of admixed Latino populations from the Population Architecture using Genomics and Epidemiology (PAGE)-Latin American and the Hispanic Community Health Study/Study of Latinos (HCHS/SOL) datasets to illustrate the improved resolution in revealing ancestry-specific fine structure across Latin American countries. Finally, using HCHS/SOL as the basis, we simulated stratified phenotypes to demonstrate the improved robustness to association testing if population structure provided by as-eGRM is utilized.

Material and methods

Expected pairwise genetic relatedness based on genealogical trees

We first briefly review the definition and construction of the eGRM, which provides the pairwise genetic relatedness with a genealogical tree.[25] Given a branch e on a genealogical tree t within an ARG G, the eGRM defines the genetic relatedness, R^t, between a pair of haplotypes i and j on a single tree as

$$R^t(i,j) = \sum_{e \epsilon E^t(i,j)} w(e)\mu(e) \quad \text{(Equation 1)}$$

$$\mu(e) = t(e)l(e)u(e) \quad \text{(Equation 2)}$$

where $E^t(i,j)$ denotes the set of the branches connecting haplotype i to haplotype j on tree t and $w(e)$ is a weighting function that will be discussed further below. As the number of mutations occurring on each branch e of the tree is modeled as a Poisson process, its rate is $\mu(e)$, which is the product of $t(e)$, $l(e)$, and $u(e)$, denoting the length of branch e in generations, the number of base pairs that the tree t covers, and the mutation rate on this branch, respectively.

We use $x(e)$ to denote the haplotype vector (vector of haploid individuals) associated with branch e; that is,

$$x_i(e) = \begin{cases} 1, \textit{if sample i is a descendant of } e \\ 0, \textit{otherwise} \end{cases}, 1 \le i \le N. \quad \text{(Equation 3)}$$

To computationally implement R^t, we traverse each branch e, compute $w(e)\mu(e)$, and add $w(e)\mu(e)$ to the elements in R^t indexed by the descendant samples of branch e:

$$R^t = R^t + x(e)x(e)^T w(e)\mu(e). \quad \text{(Equation 4)}$$

Therefore, across all haplotypes,

$$R^t = \sum_{e \in E^t} x(e)x(e)^T w(e)\mu(e). \quad \text{(Equation 5)}$$

Finally, the relatedness measure is averaged across all trees in the ARG, G. With centering, the eGRM is finally defined as

$$eGRM := C_N \left(\frac{1}{\mu(G)} \sum_{t \in G} R^t \right) C_N$$
$$= C_N \left(\frac{1}{\mu(G)} \sum_{e \in G} x(e)x(e)^T w(e)\mu(e) \right) C_N,$$

where $\mu(G) = \sum_{e \in G} \mu(e)$, $C_N = I_N - \frac{1}{N}\mathbf{1}\mathbf{1}^T$ is a centering matrix, and I_N is the $N \times N$ identity matrix.

Expected pairwise genetic relatedness with ancestry-specific genealogical trees

We define as-eGRM as the eGRM computed on ancestry-specific trees within G (Figure 1A). By intersecting with the local ancestry information, we prune haplotypes from the tree t that are not from the ancestry of interest and re-define the tree while setting those haplotypes as missing. In other words,

$$R^t(i,j) = \begin{cases} \sum_{e \epsilon E^t(i,j)} w(e)\mu(e), \textit{both i and j are not pruned} \\ \textit{missing, otherwise} \end{cases}. \quad \text{(Equation 6)}$$

We denote the summing matrix across the ARG G as $R^G = \sum_{t \in G} R^t$. As each tree has different number of haplotypes set as missing due to deriving its local ancestry from non-targeted ancestries, instead of dividing the summing matrix by a constant $\mu(G)$, we divide the summing matrix by a $N \times N$ matrix (denoted D^G) to account for the differential missing level while taking into account the expected number of mutations occurring on each tree (Figure 1A). Each element $D^G(i,j)$ represents the sum of non-missing $\mu(e)$ at position (i,j) across the R^t ($1 \le t \le |G|$, where $|G|$ is the number of trees in G):

$$R^G(i,j) = \frac{1}{D^G(i,j)} \sum_{t \in G} R^t(i,j) \quad \text{(Equation 7)}$$

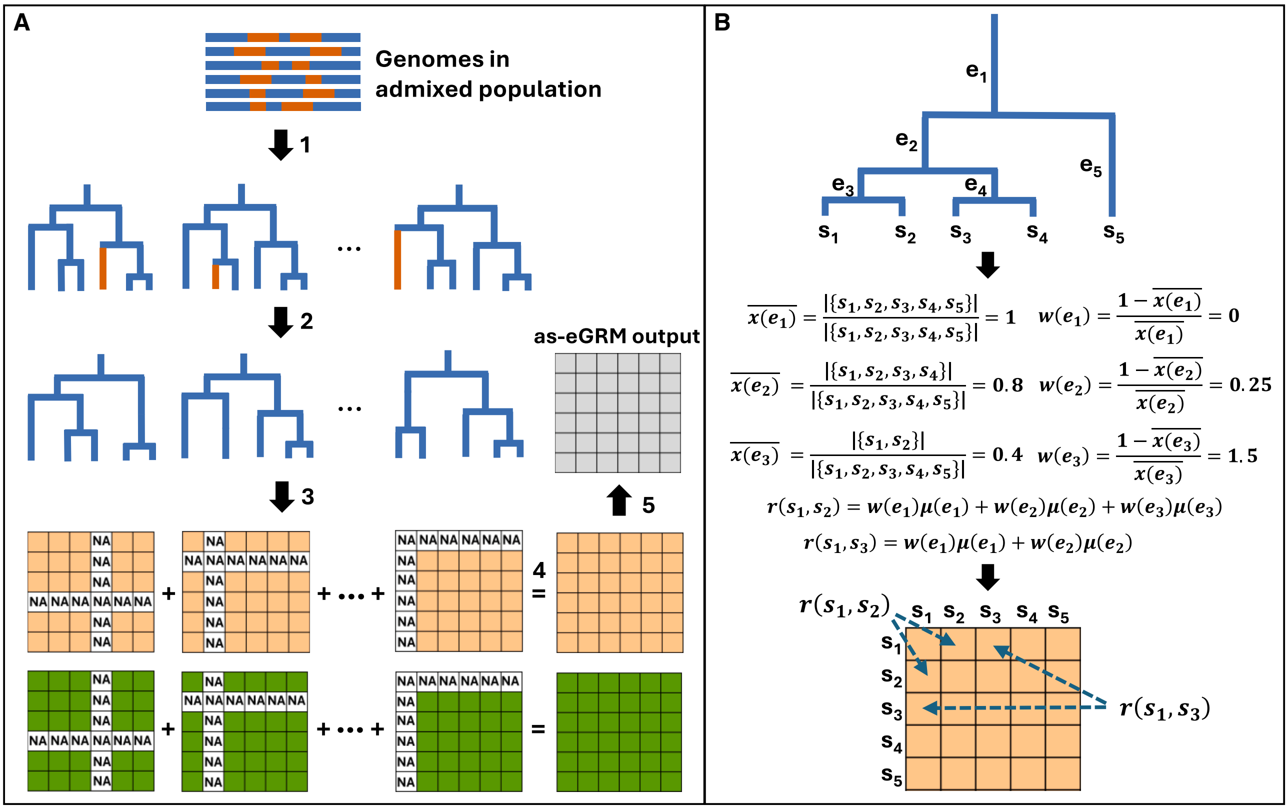

Figure 1. Design of as-eGRM

(A) A visual schematic of the implementation of as-eGRM. Tree branches colored in orange denote the genomic segments derived from the non-target ancestry and thus are removed from the trees. See the text for detailed description.

(B) A toy example of the computation of pairwise genetic relatedness. The details are described in the material and methods, Equation 5 under the section Expected pairwise genetic relatedness based on genealogical trees. Here we show a tree with five haplotypes, s_1–s_5, connected through a tree with five branches, e_1–e_5. $r(s_i, s_j)$ denotes the relatedness between s_i and s_j, $\mu(e_i)$ denotes the expected number of mutations occurring on branch e_i, and $\overline{x(e_i)}$ denotes the proportion of the descendant samples under branch e_i in all the samples. Weights on each branch, $w(e_i)$, are calculated based on $\overline{x(e_i)}$, and, given the weights and the expected number of mutations, we can compute $r(s_i, s_j)$ using all branches that connect the two haplotypes.

$$D^G(i,j) = \sum_{t \in G} \tau^t(i,j) \qquad \text{(Equation 8)}$$

$$\tau^t(i,j) = \begin{cases} \sum_{e \in E^t(i,j)} \mu(e), \textit{if both i and j are not removed} \\ 0, \textit{otherwise} \end{cases}. \qquad \text{(Equation 9)}$$

Finally, we center R^G as we would a regular eGRM:

$$as - eGRM := C_N \left(\frac{1}{D^G} \sum_{t \in G} \sum_{e \in E^t} w(e) x(e) x(e)^T \mu(e) \right) C_N. \qquad \text{(Equation 10)}$$

Choosing the weighting to better reveal recent population structure

The weights on each branch, $w(e)$, were originally defined in eGRM as $w_1(e) = \frac{1}{\overline{x(e)}(1-\overline{x(e)})}$, where $\overline{x(e)}$ denotes the proportion of the haplotypes under branch e. Therefore, this is analogous to the weights used in the canonical GRM, $\frac{1}{p(1-p)}$, to adjust for the binomial variance of variants with frequency p.[25] We found that, in the context of a genealogical tree, this weight places higher weights on both recent branches (e.g., when $\overline{x(e)}$ is small, near the leaves of the tree) as well as ancient branches (e.g., when $\overline{x(e)}$ is large, near the root of the tree; Figure S1A), since the formulation for the canonical GRM places greater weights for SNPs with rare derived alleles (younger and on branches nearer the leaves of the tree) as well as SNPs with rare ancestral alleles (older and on branches nearer the root of the tree). Because population structures in humans are likely established more recently, weighting both recent branches and older branches similarly is suboptimal; older branches (and analogously older variants that tend to have lower ancestral allele frequency) would not be informative for recent population structure. Indeed, in a simple two-subpopulation two-way admixture model (Figure S2), we observe that $\overline{x(e)}$ tends to be large for ancient branches, and small for recent branches (Figure S1C). As a result, the original weight, $w_1(e)$, tends to place higher weights on the more ancient branches, particularly when taking into account the longer branches and opportunities for mutations in those branches (Figure S1D). Motivated by this finding, we reasoned that we need a weight that breaks this symmetry with respect to the ages of the branches (and alleles) and experimented with different parametric weighting functions (Figure S3). We decided to use weighting function of the form $w_2(e) = \frac{1-\overline{x(e)}}{\overline{x(e)}}$ to be more effective in up-weighting the more recent branches nearing the leaves of the genealogical tree (Figures S1B and S1E) when computing the expected pairwise relatedness. The software as-eGRM (https://github.com/jitang-github/asegrm) allows users to input different functional forms of the weight.

Simulation of admixed populations

Three demographic models were used to simulate admixed populations: (1) a two-population split two-way admixture model, (2) a grid-like 3×3 stepping-stone model, and (3) a three-way admixed Latino model. For all models we used msprime (version 1.2.0)[26] to simulate genetic data with the recombination and mutation rates were set to 1e−8 per generation per base pair, a ploidy of 2, and 500 haplotypes (each spanning 100 Mb) per population. In the two-population split, two-way admixture model (Figure S2), and the grid-like stepping-stone model (see below), the effective population size was set to 10,000 for all populations, with no recent growth. A three-way admixed Latino model (see below) was modified from a previously published model that fitted the admixture history of self-reported Latino Americans from Los Angeles[27]; here we included substructure in this model. A visual representation and detailed parameter specifications are shown in the respective figures and supplementary figures. The commands for simulations are released with the as-eGRM software.

Quality control of empirical data

We tested as-eGRM and compared it to alternative approaches on empirical data from the HCHS/SOL and PAGE global reference datasets. The HCHS/SOL datasets were obtained from dbGAP (dbGAP: phs000880.v1.p1 and phs000810.v1.p1). HCHS/SOL is a large US-based study of 16,415 Hispanic/Latino individuals, among whom 12,803 consented to genetic studies and were successfully genotyped on a genome-wide SNP array.[28] Quality control of genotypes was performed using PLINK,[29] excluding variants that had a call rate <99% or *p* value for Hardy-Weinberg equilibrium $<1.0 \times 10^{-6}$, as well as individuals that had >2% missingness. For the HCHS/SOL data, we retained only individuals whose four grandparents were self-reported to be from the same country and filtered out relatives by removing one individual in the pairs with kinship (calculated by PLINK) greater than 0.0884 (corresponding to second-level relatives or closer). After quality-control filtering, we retained 2,036,821 variants and 8,260 individuals for analysis. The ancestry profiles systematically differed between individuals with grandparental origin from mainland/continental Latin American countries (with greater Indigenous American, or IA, ancestry) and Caribbean Latin American countries (with greater African ancestry). In this study we focused on individuals from mainland countries, including Argentina, Colombia, Costa Rica, Ecuador, El Salvador, Guatemala, Honduras, Mexico, Nicaragua, and Peru; mainland countries were removed from analysis if fewer than 15 individuals were available. Among the 8,260 individuals, 1,868 are with grandparental country memberships from these mainland countries and have an estimated IA ancestry proportion greater than 0.5 (Table S1). We analyzed this subset to be consistent with the filtering that previous study used.[6] Additionally, we also analyzed 1,671 individuals from the Chicago recruitment site with the same grandparental country memberships but across the entire ancestry proportion spectrum, to illustrate the robustness of as-eGRM to missing data (Table S1). In this analysis we chose to focus only on the Chicago recruitment site to keep the analysis scale ($\sim$1,600 individuals) comparable with that of the previous comparison in literature ($\sim$1,800 individuals).[6] The PAGE global reference dataset was obtained from dbGAP (dbGAP: phs001033.v1.p1). We extracted a subset of 630 Latin America individuals (from Peru, Venezuela, Mexico, Colombia, and Brazil) from the global reference dataset and applied the same quality-control filtering to retain 1,399,468 variants for analysis. HCHS/SOL and PAGE-Latin American data were combined with the ancestry reference (see below) and together phased by EAGLE.[30] Research and access to data through dbGaP were approved by the institutional review board at University of Southern California.

Inference of ancestral recombination graphs and local ancestry calls

We used Relate (version 1.2.0)[31] to infer ARGs for both simulated and empirical datasets. For simulated data, recombination rate, mutation rate, and effective population size were set to match the simulation parameters. For the HCHS/SOL and PAGE-Latin American data, mutation rate and effective population size were set to the default values as suggested in the user manual, along with the HapMap Phase II genetic map[32] in hg38. For computational scalability when inferring the ARG on empirical datasets, we applied Relate on chunks of 10,000 SNPs in parallel. The utility *RelateFileFormats –mode ConvertToTreeSequence* was used to convert Relate's output to the *tskit*[33] format.

RFMix (version 2)[34] was used to infer local ancestry segments in both simulated and empirical datasets. The ancestral references used in simulation are indicated in each respective simulation model. For running RFMix on the HCHS/SOL and PAGE-Latin American data, we used previously selected individuals based on gnomAD v3.1[35] as the reference. In gnomAD's nomenclature, we included 671 non-Finnish European (NFE) individuals for European ancestry, 708 African/African American (AFR) individuals for African ancestry, and 94 Admixed American (AMR) individuals (7 Colombian; 12 Karitianan; 14 Mayan; 4 Mexican in Los Angeles; 37 Peruvian in Lima, Peru; 12 Pima; and 8 Surui) for IA ancestry. Note that, while these labels follow gnomAD's nomenclature, the reference samples included only individuals from continental Europe and Africa that were labeled NFE and AFR in gnomAD, and only AMR individuals with estimated IA ancestry >85%[36,37] to lessen the impact of admixture in individuals used as ancestry references.

Implementation of previous methods to investigate population structure

We compared PCA + UMAP on the as-eGRM to that based on the canonical GRM, the original eGRM, as well as Browning's AS-MDS and mdPCA. For PCA on the canonical GRM, we pruned sites with minor-allele frequency (MAF) <0.01 and those in high linkage disequilibrium (LD) using PLINK with the command "–maf 0.01 –indep-pairwise 50 5 0.2." Then a variance-standardized GRM was computed on the pruned genotypes, followed by eigen-decomposition to derive PCs. For PCA on the eGRM, eGRM was constructed using the software package from its GitHub page (see web resources), using the same ARG input as the as-eGRM. Eigen-decomposition was performed on the output of eGRM to compute PCs. For Browning's AS-MDS and mdPCA, codes were downloaded from their respective websites (see web resources) and executed per instructions from the user manuals. We applied the same MAF and LD pruning as in PCA of the canonical GRM. In particular, mdPCA proposed five different methods (methods 1–5) for generating ancestry-specific PCs. All five methods were tested, and we found generally the best results were based on method 1, which were presented in this study. Methods that leverage local ancestry segments (Browning's AS-MDS, mdPCA, and as-eGRM) used the same local ancestry calls. Unless otherwise noted, UMAP was applied to the top 20 and 50 PCs from simulated and empirical data, respectively, using the Python package *umap* with default parameters (n_neighbors:15, min_dist:0.1, metric: Euclidean).

We compare the PCA or PCA + UMAP results based on each method, generally plotting the top two components in a biplot. We used two ways to quantify the degree of clustering effectiveness. First, we followed previous study and used the separation index (SI), which assesses the proportion of nearest neighbors that are in the same population in multidimensional space.[25] Intuitively, for each individual in a cluster of true size n, we compute the proportion of the n closest neighbor in the multidimensional space that are in the same cluster and average the proportion over all individuals in the dataset. SI is a real number between 0 and 1, indicating how well a multidimensional metric is capturing the true classification. Second, we performed K-means clustering on the PCA or PCA + UMAP (after rescaling each axis to be on the same scale) to generate clusters and used the adjusted mutual information (AMI) score[38] as implemented in scikit-learn (version 1.3.2) to measure the agreement between the clusters and the true label. Intuitively, AMI measures how much knowing one set of clustering result reduces uncertainty about another set of clustering results based on their joint probability distribution, while correcting for the expected information that would be shared by random chance. This creates a normalized measure that allows fair comparison between two sets of clustering. AMI is a real number between 0 and 1, indicating how much the clusters in one partition correspond to clusters in another partition beyond what would be expected by random chance. For both SI and AMI, the true label is the deme or population membership of each individual in simulations. In empirical data, the self-reported country of origin based on grandparental birthplaces in HCHS/SOL or the provided country of origin for PAGE global reference were used as the true label.

Evaluation of as-eGRM to improve robustness in genome-wide association studies

We used simulated non-heritable phenotype to evaluate the degree to which PCs based on the as-eGRM (and the alternative methods) can control the inflation of test statistics in a GWAS. We focused on the 4,527 HCSH/SOL individuals across all recruitment sites with grandparental country memberships from the mainland Latin American countries mentioned above to boost sample size and power in association testing (Table S1). To simulate non-heritable phenotypes, we followed a previously described approach.[39] Specifically, we simulated non-heritable phenotype y_{ij} of an individual i from country j as $y_{ij} \sim N(\mu_j, 1)$, where μ_j is the mean environmental effect in country j. We set $\mu_j = 2$ for one Latin American country (with $n > 100$) at a time and zero for all others. We then tested the non-heritable phenotypes for association with imputed genotypes across chromosomes 1–3 from the 4,527 individuals. Imputation was performed based on 368,562 array SNPs using the TOPMed imputation server,[40] and only SNPs with the imputation quality metric Rsq greater than 0.3 were retained (15,627,833 SNPs), of which only those with MAF >0.001 (3,508,560 SNPs) were tested for association. Association testing was done using PLINK (version 1.90b7) with the *–linear* function, using 10 PCs or components derived from as-eGRM and alternative approaches (canonical GRM, eGRM, mdPCA, and AS-MDS) as the covariates to test the associations. Inflation was evaluated by deviation of the test statistics at the 99th percentile, expressed as the ratio of the observed test statistic over the expected test statistic, which we denote as $\lambda_{0.99}$. Because the simulated phenotypes have no genetic contribution, any inflation of test statistics is due to uncontrolled false-positive associations. Therefore, smaller values of $\lambda_{0.99}$ correspond to greater efficacy of controlling for population stratification by a method. We also performed the same analysis grouping countries by geographic locations to increase sample size. These groupings include the northern part of Central America (Mexico; this is equivalent to the analysis at the country level as Mexico had the largest number of individuals; $n = 3{,}139$), southern part of Central America (combining Costa Rica, El Salvador, Guatemala, Honduras, and Nicaragua; $n = 900$), and South America (combining Argentina, Colombia, Ecuador, and Peru; $n = 488$). We performed each analysis in 10 independent replicates and presented the mean and standard error across the 10 replicates.

Results

An overview of the design of as-eGRM

To compute ancestry-specific expectation of genetic relatedness, we first create ancestry-specific trees from inferred genealogical trees. The mathematical formulations are described in detail in the material and methods. We intersect the inferred genealogical trees with inferred local ancestry segments (in practice inferred from existing methods such as Relate[31] and RFMix,[34] respectively; Figure 1A, step 1). We remove the leaf nodes derived from non-target ancestral populations (the orange leaf nodes in Figure 1A) to generate ancestry-specific trees (Figure 1A, step 2). Further, for each of the ancestry-specific trees, we specify two $N \times N$ matrices (named R^t and τ^t respectively; Figure 1A, step3; see material and methods). R^t (the orange matrices in Figure 1A) scores all pairwise relatedness based on the corresponding tree t in the ARG with positions indexed by one or both samples deriving ancestry from the non-target ancestries set to missing values. The pairwise relatedness is computed following the procedure illustrated in Figure 1B, which followed the principle of the original eGRM[25] that treats mutations as random, and computes the expected relatedness summed across all branches connecting the two haplotypes weighted by the probability of a mutation occurring on the branch (i.e., proportional to the branch length; material and methods). τ^t (the green matrices in Figure 1A) records the expected number of mutations on each tree corresponding to the non-missing cells in R^t, thereby tracking the differential missingness between pairs of haplotypes due to different proportions of the non-target ancestries being masked across the genome. Across all trees in the ARG, we then take the element-wise sum of the two matrices, respectively, producing two matrices R^G and D^G (Figure 1A, step 4; see material and methods). Finally, we take the element-wise ratio of the two summed matrices followed by mean-centering to generate the final as-eGRM (Figure 1A, step 5).

When computing the expectation of relatedness per branch, the original formulation from the eGRM included a weight based on the inverse of the binomial variance (see material and methods). This stemmed from the practice in the canonical GRM in which the contribution from each variant is normalized to adjust for the binomial variance of variants across different allele frequencies. In other words, alleles with extremely low and high derived allele frequencies, corresponding to alleles that tend to be very young or very old, respectively, in the sample, will be up-weighted because of their low MAFs. The conceptual analog in the case of branches in a genealogical tree is that the (young) branches near the leaves and the (old) branches near the root will be up-weighted in the eGRM (Figure S1A). We reasoned that this practice would negatively impact the ability of the eGRM to discern population structure. Structures in humans (and in most species in general) are likely established toward the leaves of the tree, perhaps within the last several hundred generations compared to the coalescent history of the sample, and thus ancient alleles or branches on the tree pre-dating the structure of interests will likely carry little information and instead contribute to the relatedness shared across all individuals.[25,39] Indeed, we observed in simulations of a simple two sub-population two-way admixture model (Figure S2) that branches connecting two individuals from the same subpopulations tend to be much more recent than branches shared by the two subpopulations (Figure 2A). However, in the original weight formulation, $w_1(e) = \frac{1}{\overline{x(e)}(1-\overline{x(e)})}$, these branches are not up-weighted compared to branches connecting individuals across subpopulations, particularly after accounting for the expected number of mutations on these branches (Figure 2B). We thus experimented with different parametric weighting functions (Figure S3) and opted to use the weights of the form $w_2(e) = \frac{1-\overline{x(e)}}{\overline{x(e)}}$ to be effective in up-weighting the more recent past of the genealogical tree (Figure 2C). Indeed, the as-eGRM using the updated weight shows clearer contrast between individuals from different subpopulations compared to as-eGRM using the original weights, resulting in clearer demarcation of the two subpopulations on PCA of the as-eGRM (Figure 2D).

as-eGRM outperforms alternative methods for delineating population structure in simulation

We used a two-split two-way admixed demographic model to simulate an admixed population with structure for evaluating the performance of as-eGRM in revealing fine-scale structure (Figure 3A). In this model, there is a first population split 2,000 generations ago, separating the orange ancestry (anc2) from the blue ancestry. A second split then occurred at 100 generations ago, creating anc1 population as well as a 3×3 stepping-stone model with bi-directional migration with rate 0.01 with neighboring demes to establish a grid-like spatial structure. Finally, 20 generations ago, there is a single pulse admixture from anc2 to the nine demes, with varying proportions (Figure 3B). We assessed the performance of as-eGRM using the SI and the AMI score (material and methods); both measures are between 0 and 1, and the higher the value the better the method is at differentiating subclusters relative to the ground truth. When we applied PCA + UMAP to the canonical GRM from the simulated data, we observed the appearance of approximately nine demes, although there are clear misclassifications of individuals that are driven by similar ancestry proportions (Figures 3C and S4; $r = -0.43$ and -0.54 between ancestry proportions and UMAP1 and UMAP2, respectively). When PCA + UMAP was applied to the eGRM without taking into account local ancestry information, there is again little power to differentiate the structure specific to the blue ancestry (anc1; SI = 0.21, AMI = 0.17; Figure 3C). While UMAP applied to the result of AS-MDS or mdPCA showed some improvement (both SI and AMI = 0.35–0.38) over the result from eGRM, the resolution is limited (Figure 3C). In contrast, as-eGRM was able to clearly delineate the nine demes, completely free from the influence of admixture from anc2 (Figure 3C). While applying UMAP on top of PCA appears to accentuate the clustering, generally similar patterns are also observed if we only examine PCA results (Figure S5, middle column), where results based on as-eGRM show the best performance (SI = 0.83, AMI = 0.9).

We also investigated the impact of different admixture proportions from anc2 as well as the timing of the split to establish the 3×3 grid structure on each method's ability to discern population structure. Greater admixture from the non-target ancestry would reduce the portion of the genome that is informative for fine-scale structure in the ancestry of interest, and more recent structure would also mean less differentiation among the demes, making fine-scale structure less discernable. We thus conducted additional simulations and evaluations with setting the admixture proportions, m1–m9, to ~0.2–0.4 and ~0.4–0.6 and setting the time of onset of structure, t_split, to 50 and 300, separately. As expected, the performance for AS-MDS and mdPCA decreased with increasing admixture proportions from anc2 (Figures S5 and S6) or more recent onset of the grid-like structure (Figures S7 and S8) both visually in biplots and by SI or AMI. The performance for both ancestry-specific approaches also improved when admixture proportions from anc2 decreased or when the grid-like structure persisted for longer (Figures S6 and S8; SI = 0.74–0.86). In all scenarios, as-eGRM consistently outperformed the alternatives, with near-perfect delineation of the nine demes, particularly when UMAP was applied on the top 20 PCs (Figures S6 and S8). The consistently poor performance of eGRM across scenarios highlights the benefits of the modifications implemented in as-eGRM.

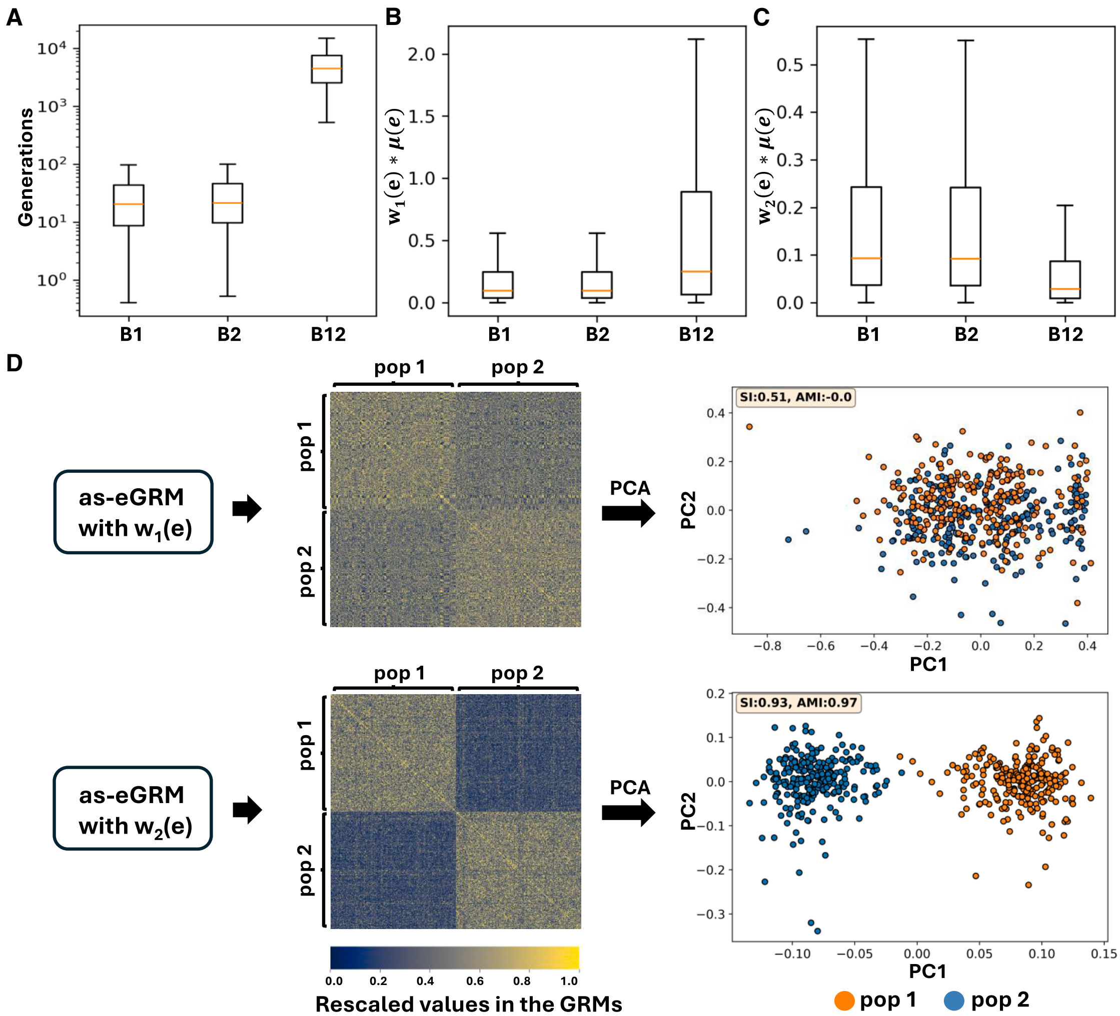

Figure 2. Up-weighting recent branches enhances as-eGRM performance in revealing finer-scale structure in admixed populations

An admixed population with a two-subpopulation (labeled pop1 and pop2) structure was simulated using the model in Figure S2.
(A) Population-common branches are more ancient than the population-specific branches. B1, B2, and B12 represent branches specific to pop1, pop2, and common to both, respectively. Population-specific branches and population-common branches are computationally defined as the branches with more than 80% of the descendants coming from one population (i.e., pop1 or pop2), and the branches with the descendants cover more than 40% of the individuals from each of pop1 and pop2, respectively.
(B) $w_1(e)$ denotes the weighting function used by eGRM. When $\mu(e)$ is weighted by $w_1(e)$, population-specific branches are not up-weighted relative to the population-common branches.
(C) $w_2(e)$ denotes the current weighting function used by as-eGRM. When weighted by $w_2(e)$, population-specific branches are up-weighted when computing the expected relatedness between pairs of individuals because of the greater weight placed on recent branches.
(D) as-eGRM resulting from the two different weighting functions show different levels of contrast between individuals from each of the subpopulations. as-eGRM using $w_2(e)$ results in intra-population relatedness values that are significantly higher than inter-population values, facilitating PCA-based population separation. The as-eGRMs were visualized as heatmaps. To aid in visualization, we rescaled the middle 90% of the as-eGRM values to be within range of 0–1 and set the outlier to the boundary values. PCA was applied to the original, untransformed as-eGRM.

We further evaluated as-eGRM on a more realistic three-way admixed Latino demographic history previously fitted from the inferred genealogical trees from array genetic data of Latinos residing in Los Angeles, California.[27] We modified this model to include recent population split at 50, 100, or 300 generations ago (Figure 4A). Both subpopulations received the same amount of introgression from two other ancestries (10.7% from an “African-like” ancestry and 44.2% from a “European-like” ancestry) at 25 generations ago (Figure 4B). Again, as-eGRM tended to outperform alternatives in discerning the population structure in PCA (Figure 4C), including scenarios with more recent structure (Figures S9 and S10). In this more realistic scenario, the canonical GRM, AS-MDS, and mdPCA generally had difficulties delineating the two subclusters (Figures 4C, S9, and S10).

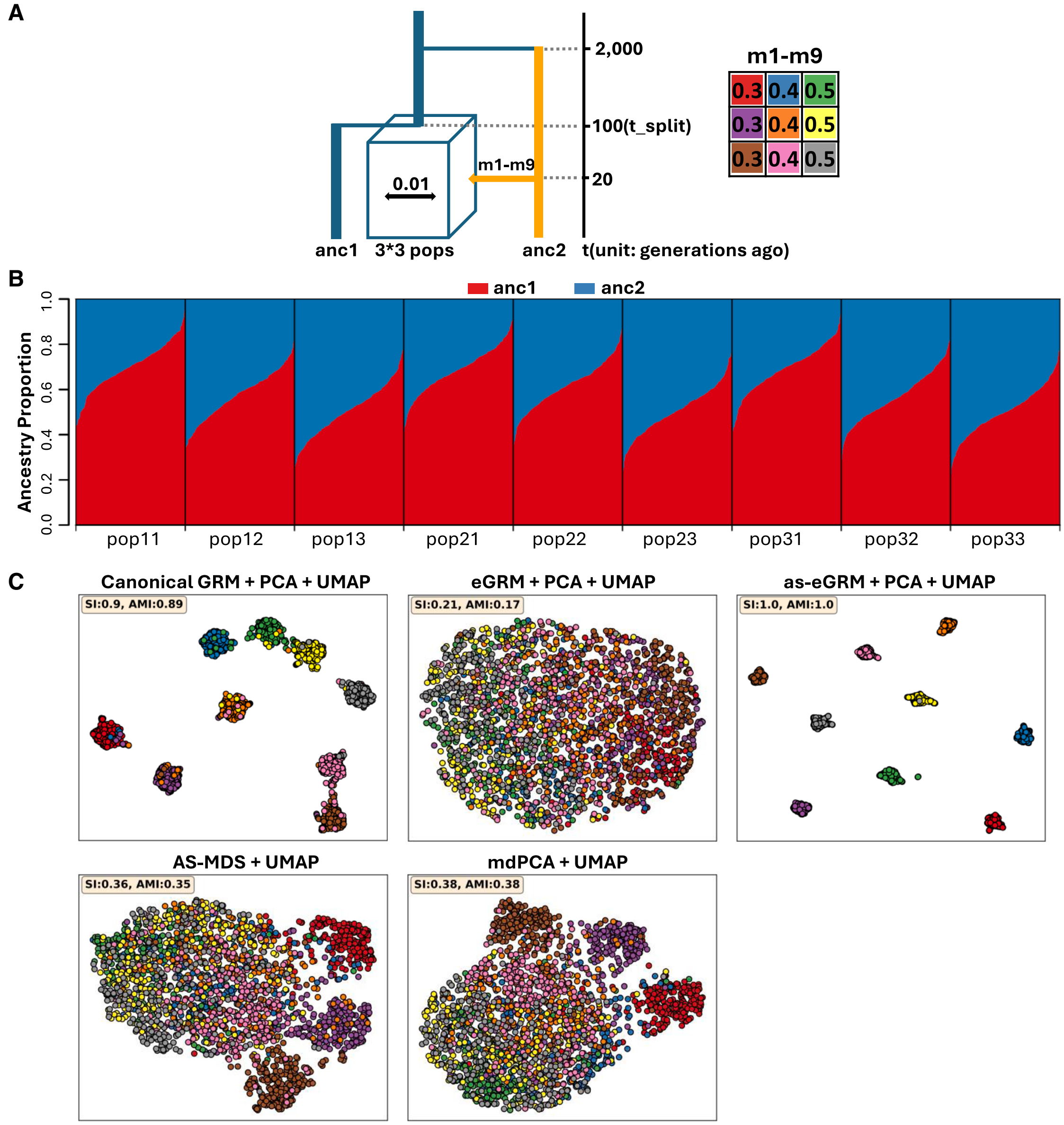

Figure 3. as-eGRM outperforms alternative methods when applied to an admixed population with a grid-like spatial structure

(A) The demographic model for simulating an admixed population with a 3 × 3 grid-like subpopulations structure. *anc1* and *anc2* represent ancestral populations and were used as the reference for local ancestry inference. *m1–m9* specify the proportions of genomic components derived from *anc2* for the individuals in the nine demes, respectively. *t_split* and *t_admix* specify the time the nine subpopulations split and the time the admixture event happened, respectively. Recent migration (rate: 0.01) between neighboring demes has occurred over the last 10 generations.

(B) Ancestry proportions of the individuals in the nine subpopulations, as inferred by RFMix.

(C) The performance of PCA followed by UMAP applied to the canonical GRM, the eGRM, the as-eGRM, as well as UMAP applied to AS-MDS and mdPCA. 20 PCs were projected down to two dimensions by UMAP, as shown in biplots. Data points represent individuals, with colors indicating population membership. Axes for UMAP plots are not labeled as distances are meaningless after UMAP transformation.

as-eGRM better delineates population structure in Latin America

We applied as-eGRM to genotyping array data of individuals from Latin America to evaluate its ability to delineate fine-scale population structure in empirical analysis. Many of the Latino individuals have admixed genomes consisting of three predominant continental ancestries: Indigenous American (IA; primarily of South and Central America, Mexico, and the Caribbean islands), European as a result of colonization, and African as a result of slave transport from West Africa.[5] We focused on visualizing the fine-scale structure from the IA ancestry component.

We first examined the Latino populations from the PAGE study.[41] We took the recorded country of origin as the

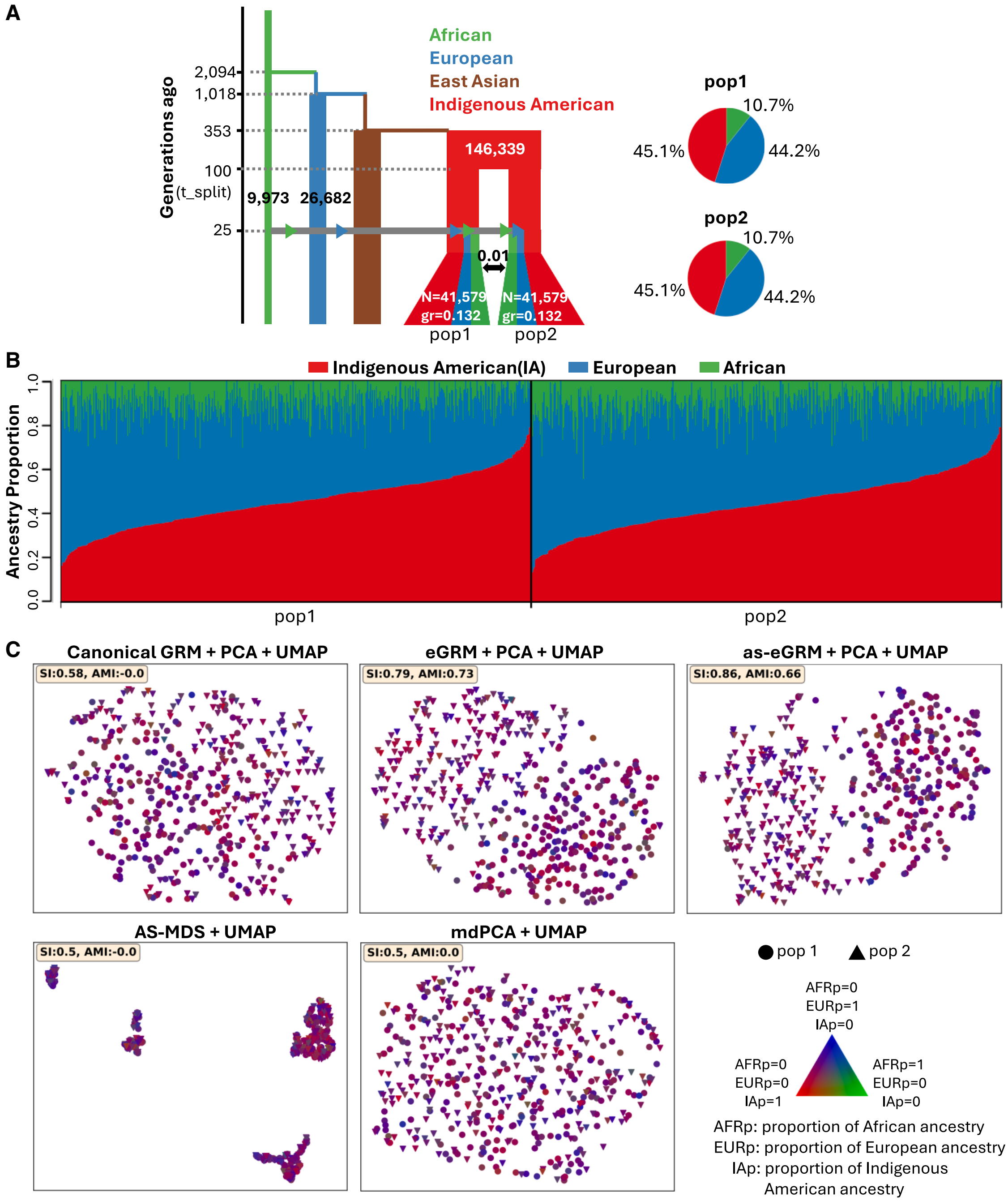

Figure 4. as-eGRM outperforms alternative methods when applied to simulated Latino populations
(A) The demographic model for simulating a Latino population with a two-subpopulation structure. Recent migration (rate: 0.01) between the two subpopulations has occurred over the last 10 generations. The model is adapted from Fan et al.[27] The ancestral populations African, European, and Indigenous American were used as the reference for local ancestry inference.
(B) The ancestry proportions of the individuals in the two subpopulations, as inferred by RFMix.
(C) The performance of PCA followed by UMAP applied to the canonical GRM, eGRM, and as-eGRM, and UMAP applied to AS-MDS and mdPCA, on revealing the two-subpopulation structure. 10 PCs were projected down to two dimensions by UMAP, shown as biplots. Data points represent individuals, with shape and color denoting population membership and ancestry proportions, respectively, as annotated in the lower right corner. In this figure, t_split = 100 was used in simulation; see Figure S10 for the scenarios where t_split = 50 or 300. Axes for UMAP plots are not labeled as distances are meaningless after UMAP transformation.

truth, hypothesizing that different countries across the Central and South America will be correlated with the fine-scale structure within the IA ancestry component. We found that PCA or PCA followed by UMAP approaches generally can discern the population structure in this data-set, although PCA based on the canonical GRM or the eGRM appears to be driven by ancestry proportions (Figure S11, left column; r = −0.79 and −0.73 for PC1 of canonical GRM and eGRM, respectively; r = −0.12 and 0.21 for PC2 of canonical GRM and eGRM, respectively). All

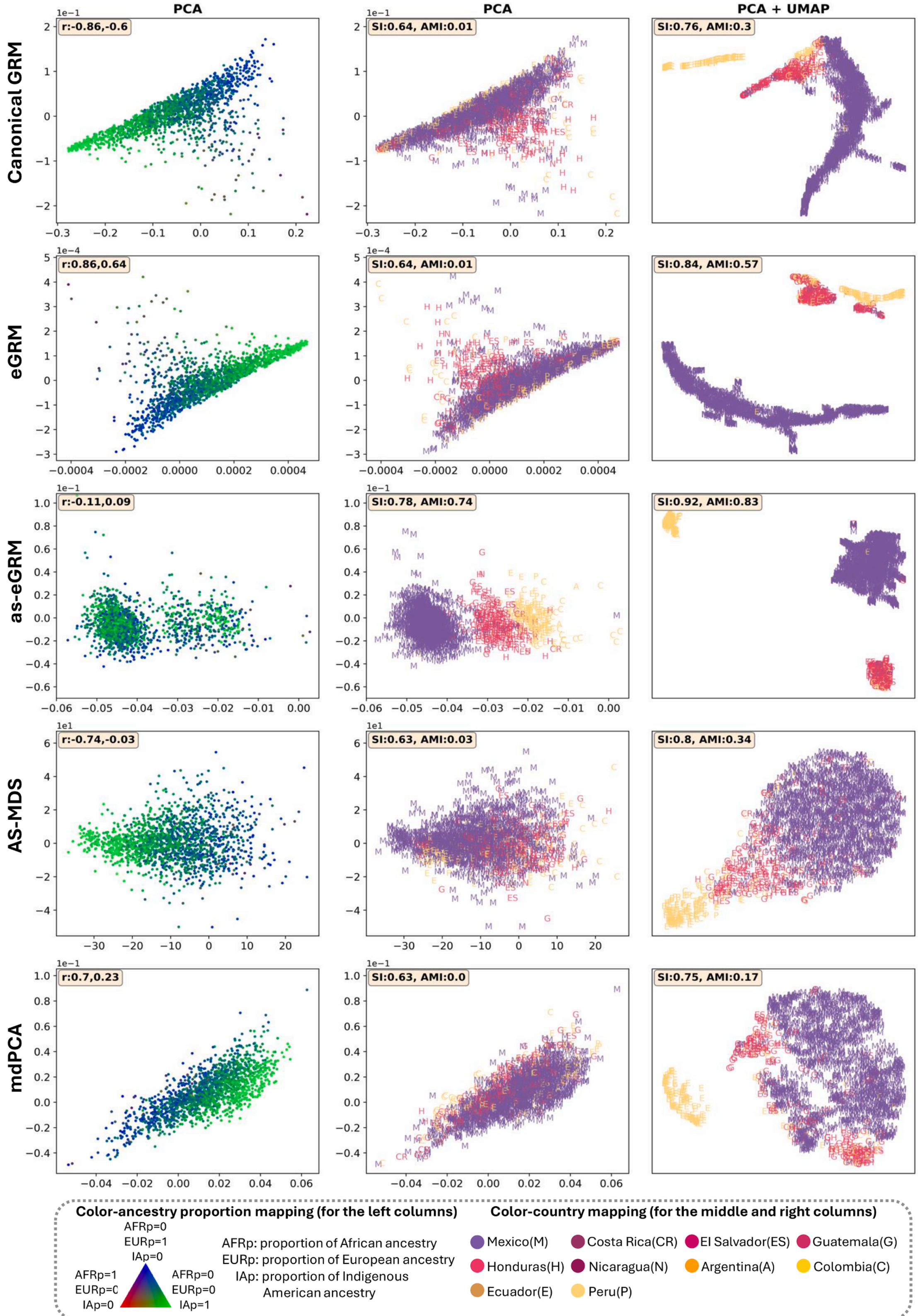

Figure 5. as-eGRM outperforms alternative methods in revealing the IA ancestry-specific structure in the Hispanic/Latino population using HCHS/SOL data

Analysis focused on 1,671 individuals from the Chicago recruitment site only but spanned the entire spectrum of ancestry proportions. Plots showed PCA or PCA + UMAP results (column annotations) of each analytical approach (row annotations). Points represent individuals, colored by ancestry proportions (left column) or grandparental country of origin (middle and right columns; see bottom box

(legend continued on next page)

ancestry-specific methods (i.e., AS-MDS, mdPCA, and as-eGRM) outperform PCA on canonical GRM and eGRM and are relatively free from bias by global ancestry (r = −0.05–0.33 across methods). Based on either the SI or AMI, as-eGRM may produce slightly more accurate clustering, but the differences are small (SI = 0.85 for as-eGRM vs. 0.81 or 0.82 for AS-MDS and mdPCA; AMI = 0.78–0.8 for the three methods; Figure S11). All methods performed similarly by SI when UMAP is applied to the PCs (Figure S11). The general ability of each method to delineate the population structure may be due to the relatively high level of IA ancestry in this sample (Figure S12). The fact that PCA based on the canonical GRM or eGRM can also somewhat elucidate the IA-specific structure suggests that the IA component across individuals in this dataset may be sufficiently differentiated and correlated with country of origin.

We then studied the Latino population from the Hispanic Community Health Study/Study of Latinos (HCHS/SOL). Previous studies applied the AS-MDS to the HCHS/SOL data and identified fine-scale structure within the IA ancestry that is consistent with grandparental country of origin.[6] Given that individuals with low levels of IA ancestry will have limited genetic data after masking, thereby adding noise to the PCA, previous studies also restricted their analysis to only individuals with at least 50% of their genomes derived from the ancestry of interest. Indeed, when we restricted our analysis to the subset of individuals with estimated IA ancestry >0.5 (across all recruitment centers), all ancestry-specific methods were able to delineate the population structure better than PCA on the canonical GRM and eGRM (SI = 0.88–0.96 vs. 0.65 and 0.67 for canonical GRM and eGRM, respectively; AMI = 0.47–0.91 vs. 0.02 and 0.11 for canonical GRM and eGRM, respectively; Figure S13, left column). When examining the distribution of IA ancestry across individuals, all methods except the as-eGRM showed substantial correlation with IA ancestry on either of the first two PCs (Figure S13, left column; |Pearson's correlation r| = 0.72–0.78). Applying UMAP on the top 50 PCs to collapse them down to two dimensions further improved the delineation of population structure for all methods (SI = 0.84–0.97, AMI = 0.42–0.91 across all methods; Figure S13, right column). Consistent with a previous report,[6] we observed clearly three to four clusters in this dataset, corresponding to Latinos from the northern part of Central America (Mexico), southern part of Central America (Costa Rica, El Salvador, Guatemala, Honduras, and Nicaragua), and southern America (Argentina, Colombia, Ecuador, and Peru).

However, when we applied each method to HCHS/SOL data spanning the entire spectrum of IA ancestry (Figure S14), the advantage from as-eGRM became apparent. In this most inclusive scenario, we found that neither the frequency-based approaches (AS-MDS and mdPCA) nor the non-ancestry-specific approach (PCA on canonical GRM and eGRM) could appropriately delineate the structure as defined by grandparental country of origin (Figure 5), which was more apparent when only analyzing the subset of individuals with high IA ancestry (Figure S13). Any pattern that was discernable from PCA was strongly correlated with global ancestry, particularly the European ancestry (Figure 5, left column; |Pearson's correlation r| = 0.7–0.86). In contrast, as-eGRM significantly outperformed all alternatives; it showed clearer separation by major grandparental country of origin in PCA, which is not confounded by proportion of global ancestry (Figure 5). Applying UMAP on the top 50 PCs somewhat improved the clustering (Figure 5, right column). However, while the expected clusters based on northern Central America, southern Central America, and South America may start to separate in analysis using the canonical GRM, eGRM, or AS-MDS and mdPCA, they are far from the clean distinct clusters when as-eGRM was used, as evident by the AMI metric (AMI = 0.83 for as-eGRM vs. 0.17–0.57 for other methods).

as-eGRM improves the robustness of association tests

Incomplete correction of population structure is one of the main sources of confounding in GWASs. To demonstrate that a better understanding of the fine-scale population structure can improve protection against confounding in GWASs, we simulated non-heritable phenotypes and tested their association to genotypes across the frequency ranges for association. Specifically, we focused on the expanded set of 4,527 individuals from HCHS/SOL for whom all four grandparents were self-reported to originate from one of the 10 mainland Latin America countries (material and methods) to increase sample size. For four of the countries with the largest samples (Mexico, n = 3,139; Honduras, n = 250; Nicaragua, n = 341; Colombia, n = 155), we in turn assigned a phenotype distribution with a shifted mean compared to the rest of the study sample. Because this phenotype is non-heritable but correlated with geographical structure, any inflation when the phenotype is tested for association with the imputed genotypes would suggest an incomplete correction of population structure. We thus examined the behavior of the test statistics in the 99th percentile tail ($\lambda_{0.99}$) as a measure of robustness when PCs from the as-eGRM or other methods are used as covariates in the association testing (material and methods). We stratified our investigation across 1,722,395 common SNPs (MAF

for annotation). The r in the left upper corner of the left column represents the Pearson correlation coefficients between the proportions of IA ancestry and PC1 (the first number) or PC2 (the second number), respectively. The separation index (SI) in the left upper corner of the right two columns is calculated using grandparental country of origin as (presumed) true labels. Axes for UMAP plots are not labeled as distances are meaningless after UMAP transformation.

between 0.01 and 0.5) and 1,786,165 rare SNPs (MAF between 0.001 and 0.01) on chromosomes 1–3 imputed from the TOPMed reference panel.[40]

When no correction is applied, we observed rampant inflation of the test statistics at the 99th percentile tail (Table S2), particularly when Honduras was the country with the localized environmental effect ($\lambda_{0.99} = 2.55$ and 2.90 for common and rare variants, respectively). For common variants, generally including 10 PCs from any of the methods tested provided decent control for population structure ($\lambda_{0.99} = 1.05$–1.18, although in some cases $\lambda_{0.99}$ can be as high as 1.44; Table S2), perhaps because common variants are expected to be shared across these countries. For the rare variants, the inflation of test statistics became more difficult to control, although as-eGRM compares favorably to other methods, particularly to AS-MDS and mdPCA. For example, when Honduras is the focal country, $\lambda_{0.99} = 1.18$ when PCs from as-eGRM were used to control for population structure while $\lambda_{0.99} = 1.74$–1.81 for AS-MDS and mdPCA (Table S2). When we grouped analysis by geographical regions so to increase the number of individuals with shifted distribution, this pattern became more obvious. When we grouped individuals from southern part of Central America ($n = 900$) or South America ($n = 488$) and performed the same evaluation, correction through PCs from as-eGRM again tended to do best in controlling for confounding, particularly for the rare variants (Table S3). For example, when Central America (south) was used as the focal population, $\lambda_{0.99} = 3.88$ when no correction was applied, 1.03 when PCs from as-eGRM were included, 1.15 when PCs from canonical GRM were included, and 1.34–1.60 when PCs from other methods were included (Table S3).

Discussion

In this study, we introduced as-eGRM, a framework that leverages genealogical trees and local ancestry information to reveal ancestry-specific structures in admixed populations. A key advancement of as-eGRM is defining ancestry-specific pairwise relatedness between individuals based on genealogical trees and local ancestry callsets across the genome accounting for missing data (due to masked non-target ancestry). Another key advancement is modifying the weighting of branches on the trees to up-weight recent branches more informative of recent population structure. Through extensive evaluation using multiple simulated and empirical datasets, we demonstrated that as-eGRM consistently outperforms alternative metrics or methods in revealing ancestry-specific population structure across various demographic scenarios and global ancestry proportions. Moreover, in an association testing setting rooted in empirical genotypes with a simulated non-heritable phenotype, we showed that corrections using PCs from the as-eGRM performed favorably compared to PCs from the alternative methods we considered, particularly for rare variants that tend to be more stratified across geography.

The original eGRM adopted the weights $w(e) = (\overline{x(e)}(1-\overline{x(e)}))^{\alpha}$, where $\overline{x(e)}$ denotes the proportion of the haplotypes under branch e and $\alpha = -1$. This is analogous to weighting by the variance of allele frequency in the formulation of the canonical GRM, aimed to up-weight rare variants (or the branches containing them) when computing genetic relatedness regardless of their ages, as rare derived alleles (contained on younger branches) or rare ancestral alleles (contained on older branches) are weighted equally. In our work, we searched for other weights that could better accentuate the recent population structure, opting for weights that upweights the recent branches (which tend to harbor younger and rarer derived alleles) than ancient branches (Figures 2 and S3). In this sense, we introduced an evolutionary concept into genetic relatedness, breaking the symmetry with respect to allele frequencies by emphasizing the age of the alleles and branches. Note that there has been extensive discussion about this weighting function when computing genetic relatedness in the context of heritability estimation. However, most of those discussions focused on choosing or estimating the appropriate α with respect to phenotypes,[42–44] but the symmetry between derived and ancestral alleles is usually maintained.

In this study, we opted to illustrate the power of our method using individuals from Latin America from the HCHS/SOL dataset as the empirical example. Latinos are known to exhibit substantial heterogeneity in the distribution of their genetic ancestry across Latin America[45,46] or even across geographical locations within the United States.[37,47] However, Latinos across geographical space tend to be aggregated for analysis. Combined with heterogeneous exposure through the environment, such aggregation has been shown to mask differences in the phenotype distribution, efficacy of polygenic risk score, and evaluation of interaction between ancestry and environment.[48] These differences can become more apparent in a stratified analysis even if just based on the self-reported ethnicity or country of origin.[48] Furthermore, the definition of the IA component of genetic ancestry tends to also aggregate putative reference individuals across Latin America, again masking the allele frequency differences that arise in ancestral Indigenous American populations across the Americas due to their unique migration history.[49,50] Indeed, previous assessment of population structure in Latin America have also shown the fine-scale differentiation within a broad umbrella term of IA ancestry.[4,51] The as-eGRM thus stands to improve these investigations leveraging the genetic linkage information. By providing a more nuanced view of genetic variation within (somewhat arbitrarily defined) ancestry components, as-eGRM helps re-define or re-interpret genetic ancestry, improves our understanding of population history, and may lead to more robust and interpretable findings in studies of diseases and complex traits. This advancement could pave the way for more precise and equitable genetic research, ultimately contributing to better health outcomes for diverse populations.

We also opted to utilize UMAP to help explore the population structure of our simulated and empirical datasets. There have been well-known recent discussions on social media regarding PCA and UMAP in the context of their applications to represent the genetic diversity of the All-of-Us cohort.[52] While the majority of the criticism centered on the conflation of genetic ancestry and self-reported race and ethnicity through controversial use of colors and labels (as described in the article "All of Us failed"; see web resources), UMAP was also suggested to contribute toward forcing a discrete nature of genetic diversity in an inherently continuous space (as visualized by PCA plots). Indeed, while the admixture process in humans is modeled as an inherently linear process, UMAP does not preserve the distances in its transformation but instead accentuates the distinctiveness of the majority subgroups. But note that both PCA and UMAP, when applied to genetic data, are dimensional-reduction approaches to reduce the high-dimensional genetic data down to visualizable two or three dimensions for exploratory analysis. The representation of the genetic data will not be loss-less through any form of dimensional reduction techniques, and the appropriate usage may depend on the context. The appropriate use of UMAP on human data is continually being explored,[53] and it may be more suitable in the context of exploring isolated islands where significant drift may occur, for instance.[54] In our context, simulated data assumed a discrete nature (e.g., the nine-deme model; Figure 3) and UMAP could be more powerful in identifying these clusters. Similarly, in our empirical application, we targeted the IA ancestry and assumed that the fine-scale structures of interest are more discrete in nature. This assumption is made whenever one operates under a generally discrete view of genetic ancestry (when distinct reference ancestral populations are presumed when modeling admixture history, for instance), even though this may or may not reflect the reality.

On a technical level, we found that population structure analysis based on the as-eGRM excels over previous methods (AS-MDS or mdPCA) when the proportion of admixture from non-target ancestry is high. For instance, as the non-target ancestry increased from 0.2–0.4 to 0.4–0.6 in the nine-deme stepping-stone model (Figure S6), mdPCA progressively performed worse in detecting population structure (SI changed from 0.74 to 0.15), while as-eGRM maintained sensitivity (Figure S6). This may also underlie the observation that AS-MDS and mdPCA performed comparably to as-eGRM on the PAGE-Latin American dataset (Figure S11; mean IA ancestry proportion = 0.68) or the HCHS/SOL data when filtering on individuals with estimated IA ancestry >0.5 (Figure S13). One reason for this observation is the impact due to missing data. As admixture proportions from non-target ancestry increase, the proportion of the genomes between a pair of individuals that are not masked by AS-MDS or mdPCA decreases, reducing the information available to compute genetic similarity between the pair. Similar issues with pervasive missingness in the GRM had been discussed in literature, particularly when using data from ultra-low-coverage sequencing data or ancient DNA (aDNA). Common approaches to deal with missingness when inferring population structure include filtering of individuals with high missingness or imputing the missingness by mean genotype values,[11,55–57] although both approaches could introduce bias in population structure inference. Other approaches, such as those based on an expectation-maximization algorithm to iteratively impute frequency of missing genotypes,[58] or based on matrix denoising techniques and truncated SVD as used by mdPCA, have also been proposed to deal with the non-random missingness in the data. In our as-eGRM framework, we did not explicitly deal with missingness in the construction of as-eGRM; we also simply ignored the regions of genome between pairs of individuals where one or both individuals have non-target ancestries. Indeed, we found that the variance in our estimates of ancestry-specific relatedness to be relatively small, oftentimes one order of magnitude lower than the estimates themselves even when missingness is around 90% (Figure S15). While our approach appears to be robust to increased admixture proportions, its ability to elucidate population structure may still suffer when investigating structure within a minor ancestry component or when ancestry segments are not randomly distributed in the genome (e.g., in the presence of adaptive introgression). We would also expect the variance of the relatedness estimates to be larger if the ARG reconstruction is less accurate or if less genetic information is available for ARG reconstruction (e.g., array genotypes were used). Therefore, future improvements may focus on evaluating and implementing approaches to ensure robustness across the spectrum of missing information.

The current implementation of as-eGRM has some limitations and future directions for improvement. First, we found that up-weighting recent branches is crucial for revealing contemporary fine-scale structures. This finding suggests that selectively weighting branches from different parts of the trees could enable the detection of structures from specific time periods. While our current approach empirically determines the weighting function for recent branches, future research should explore systematic methods to derive optimal weighting functions for both recent and temporally specific structures. Second, as-eGRM's reliance on ARG reconstruction makes it computationally intensive for datasets exceeding a few thousand individuals. ARG-reconstruction methods that can scale to biobank level data are available,[59,60] although its accuracy can still be improved.[25,61,62] We chose to use Relate as the current best combination of accuracy and scalability and also expect that rapid advances in ARG-reconstruction methods will likely improve both the accuracy and scalability, which will benefit the as-eGRM framework. Third, our method cannot yet be applied to aDNA data, as the quality of aDNA data cannot yet be used in local ancestry inference (as the target or the reference), and its incorporation into the ARG is still in development. Nevertheless, their incorporation into population structure analysis may be illuminating for both understanding the history of a modern admixed population and in interpreting or re-defining the ancestries of an admixed individual. For the time being, allelic-based approaches for incorporating aDNA may still be the most reliable. In fact, the explicit reliance of defining a high-quality reference panel and inference of discrete local ancestry labels is a strong limitation of the current approach. While it may be clearer to define ancestral populations for continentally admixed populations, the notion of ancestry is complicated by both geographical location and temporal reference.[63] Genealogical trees potentially enable a continuous view of population structure and ancestry across time, moving beyond traditional discrete ancestry classifications. Therefore, future development may also move toward a more fluid definition of ancestries and investigate the population structure at multiple levels within a cross section of time.

Data and code availability

We have implemented the algorithms related to as-eGRM with Python and C. To ease the installation, we packaged it into a Python package, asegrm, and made it publicly available in PyPI. A detailed documentation on the installation and usage is provided on its GitHub page (https://github.com/jitang-github/asegrm). The codes for reproducing the analyses in this study can also be found on the GitHub page.

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Author contributions

C.W.K.C. conceived of and designed the study. J.T. implemented the method and performed the analysis. J.T. and C.W.K.C. interpreted the data and wrote the paper.

Declaration of interests

The authors declare no competing interests.

Supplemental information

Supplemental information can be found online at https://doi.org/10.1016/j.ajhg.2025.06.016.

Web resources

All of Us failed, https://liorpachter.wordpress.com/2024/02/26/all-of-us-failed/

as-eGRM, https://github.com/jitang-github/asegrm

AS-MDS, https://faculty.washington.edu/sguy/local_ancestry_pipeline/

eGRM, https://github.com/Ephraim-usc/egrm

mdPCA, https://github.com/AI-sandbox/mdPCA

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